Enrollment No.: 240840116014

Practical-6

Aim: Create basic data visualizations (Line, Bar, Histogram, Scatter, Pie).

Step 1: Create a Line Chart import Libraries

```
#import the pandas library.
import pandas as pd

#imoprt the mutplotlib library for creating graph
import matplotlib.pyplot as plt
```

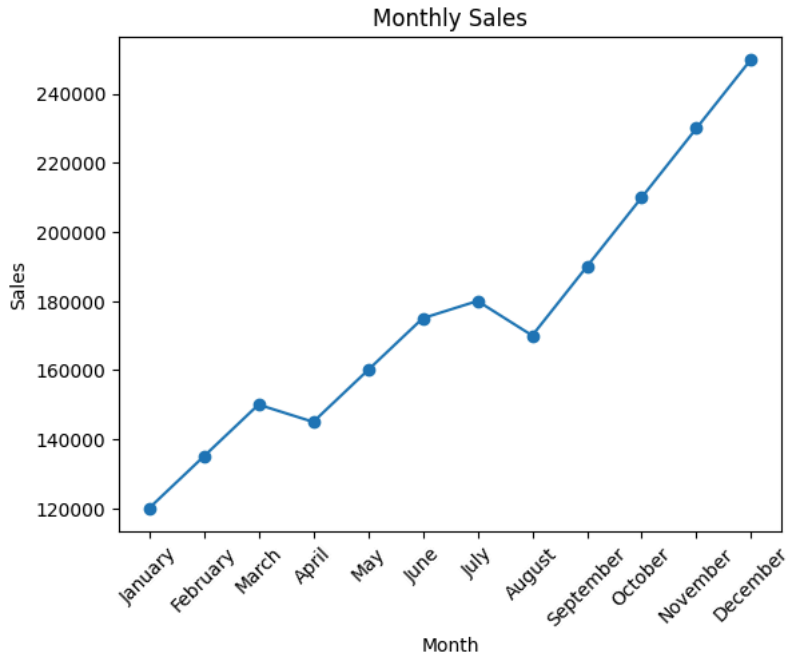
```
data ={
    "Month":['January', 'February', 'March', 'April', 'May', 'June',
             'July', 'August', 'September', 'October', 'November', 'December'],
"Product":['Laptop', 'Mobile', 'Tablet', 'Laptop', 'Mobile', 'Tablet','Laptop', 'Mobile', 'Tablet','Laptop', 'Mobile', 'Tabl
    "Region":['North', 'South', 'East', 'West', 'North', 'South', 'East', 'West', 'North', 'South', 'East', 'West'],
"Sales": [120000, 135000, 150000, 145000, 160000, 175000,
               180000, 170000, 190000, 210000, 230000, 250000],
    "Units_Sold":[12,30,25,14,35,30,18,38,32,21,45,40]
    }

df = pd.DataFrame(data)

print(df)
```

```
        Month Product Region   Sales  Units_Sold
0     January  Laptop  North  120000          12
1    February  Mobile  South  135000          30
2       March  Tablet   East  150000          25
3       April  Laptop   West  145000          14
4         May  Mobile  North  160000          35
5        June  Tablet  South  175000          30
6        July  Laptop   East  180000          18
7      August  Mobile   West  170000          38
8   September  Tablet  North  190000          32
9     October  Laptop  South  210000          21
10   November  Mobile   East  230000          45
11   December  Tablet   West  250000          40
```

```
# Line chart
plt.plot(df["Month"], df["Sales"], marker="o")
plt.title("Monthly Sales")
plt.xlabel("Month")
plt.ylabel("Sales")
plt.xticks(rotation=45)
plt.show()
```



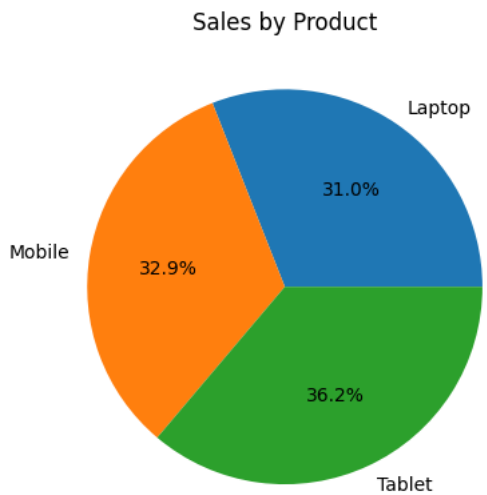
Step 2: January to December Sales

```
# Show January and December sales
print("January Sales:", df["Sales"][0])
print("December Sales:", df["Sales"][11])
```

```
January Sales: 120000
December Sales: 250000
```

Step 3: Pie Chart

```
# Product wise sales
x = df.groupby("Product")["Sales"].sum()
plt.pie(x, labels=x.index, autopct="%1.1f%%")
plt.title("Sales by Product")
plt.show()
# Show product sales
print(x)
```

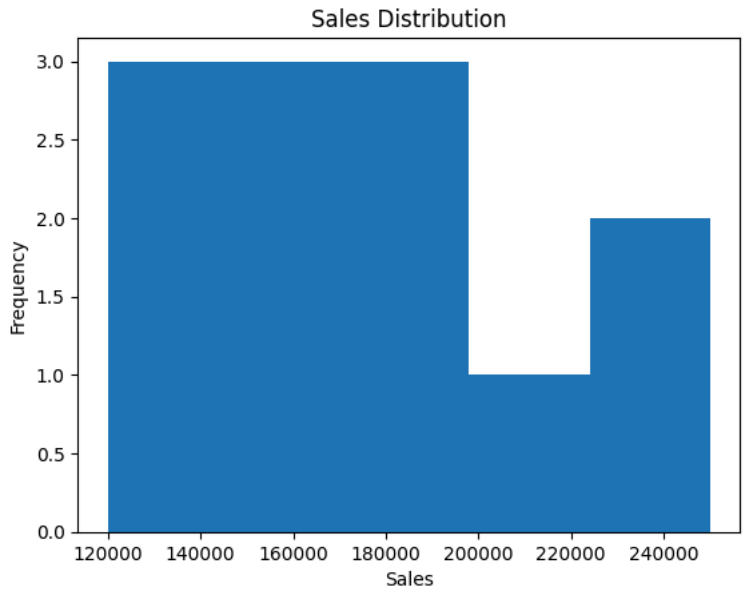



```
Product
Laptop    655000
Mobile    695000
Tablet    765000
Name: Sales, dtype: int64
```

Step 4: Create Histogram

```
# Histogram of sales
plt.hist(df["Sales"], bins=5)
plt.title("Sales Distribution")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.show()

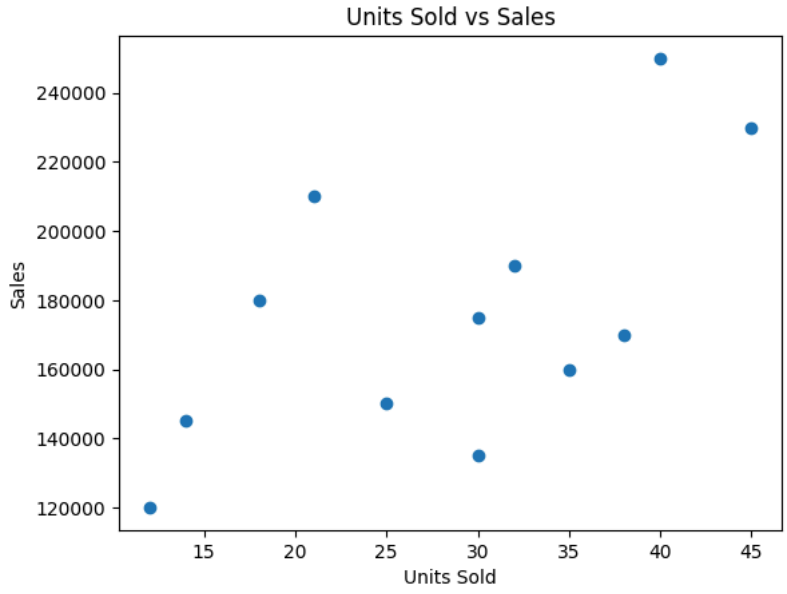
# Show sales distribution
print(df["Sales"].describe())
```




```
count        12.000000
mean     176250.000000
std       38737.167963
min      120000.000000
25%      148750.000000
50%      172500.000000
75%      195000.000000
max      250000.000000
Name: Sales, dtype: float64
```

Step 5: Units Sold and Total Sales

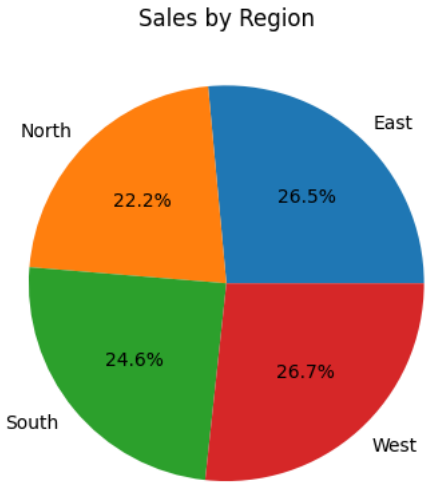
```
# Scatter plot
plt.scatter(df["Units_Sold"], df["Sales"])
plt.xlabel("Units Sold")
plt.ylabel("Sales")
plt.title("Units Sold vs Sales")
plt.show()
```



Step 6: Pie Chart – Region Sales

```
# Region wise sales
x = df.groupby("Region")["Sales"].sum()
plt.pie(x, labels=x.index, autopct="%1.1f%%")
plt.title("Sales by Region")

plt.show()
```



Step 7: Save Dataset

```
# Save data as CSV
df.to_csv("sales.csv", index=False)
print("File saved")
```

```
File saved
```

Step 8: Download CSV

```
# Download CSV file
```